

Brutal Practice Paper B33 - Nutrition in Animals | Symmetry | Forests

| Algebraic Expressions

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A person's small intestine is about 7.5 metres long while the large intestine is about 1.5 metres long. The small intestine is how many times as long as the large intestine? [\[Nutrition in Animals\]](#)
 - (A) 5.5 times
 - (B) 9 times
 - (C) 0.2 times
 - (D) 4 times
 - (E) 5 times
 - (F) 6 times
 - (G) 3 times

2. A regular hexagon can be turned about its centre and still look the same. The smallest angle of rotation that maps it onto itself is? [\[Symmetry\]](#)
 - (A) 45 deg
 - (B) 60 deg
 - (C) 72 deg
 - (D) 6 deg
 - (E) 90 deg
 - (F) 30 deg

3. A forest covers 500 hectares, and 30 percent of this area is dense canopy formed by the tallest trees. The canopy area is? [\[Forests\]](#)
 - (A) 300 ha
 - (B) 1500 ha
 - (C) 15 ha
 - (D) 50 ha
 - (E) 170 ha
 - (F) 150 ha
 - (G) 350 ha

4. Find the value of the expression $3x^2 - 2x + 5$ when x equals 4. [\[Algebraic Expressions\]](#)
 - (A) 49
 - (B) 45
 - (C) 53
 - (D) 37
 - (E) 41
 - (F) 141

5. Saliva in the mouth begins breaking starch into sugar. If 30 grams of starch is acted on and 80 percent of it is converted, the mass of starch turned to sugar is? [\[Nutrition in Animals\]](#)
- (A) 24 grams
 - (B) 6 grams
 - (C) 12 grams
 - (D) 22 grams
 - (E) 2.4 grams
 - (F) 240 grams
6. A figure looks exactly the same four times during one full turn, so it has rotational symmetry of order 4. The smallest angle that maps it onto itself is? [\[Symmetry\]](#)
- (A) 90 deg
 - (B) 45 deg
 - (C) 60 deg
 - (D) 180 deg
 - (E) 30 deg
 - (F) 4 deg
7. One mature tree releases about 100 kilograms of oxygen in a year. A grove of 20 such trees releases how much oxygen in a year, in tonnes? [\[Forests\]](#)
- (A) 0.2 tonnes
 - (B) 120 tonnes
 - (C) 20 tonnes
 - (D) 2000 tonnes
 - (E) 5 tonnes
 - (F) 2 tonnes
8. Simplify $5a + 3b - 2a + 4b$ by collecting like terms, then find its value when a equals 2 and b equals 1. [\[Algebraic Expressions\]](#)
- (A) 13
 - (B) 23
 - (C) 12
 - (D) 9
 - (E) 16
 - (F) 19
9. A healthy person's mouth makes about 1.5 litres of saliva each day. Over one full week, how many millilitres of saliva is that in total? [\[Nutrition in Animals\]](#)
- (A) 1050 mL
 - (B) 10.5 mL
 - (C) 105000 mL
 - (D) 9000 mL
 - (E) 7500 mL
 - (F) 1500 mL
 - (G) 10500 mL

10. A regular octagon has the same number of lines of symmetry as it has sides. How many lines of symmetry does it have? [\[Symmetry\]](#)

- (A) 6
- (B) 8
- (C) 1
- (D) 2
- (E) 16
- (F) 4
- (G) 10

11. Decomposers on the forest floor turn fallen leaves into humus. If 40 kilograms of leaf litter falls and 75 percent of it becomes humus, the mass of humus formed is? [\[Forests\]](#)

- (A) 30 kg
- (B) 3 kg
- (C) 75 kg
- (D) 10 kg
- (E) 35 kg
- (F) 20 kg
- (G) 300 kg

12. In the expression 7 minus 4x plus x squared, what is the coefficient of x (the term with x to the first power)? [\[Algebraic Expressions\]](#)

- (A) 4
- (B) 1
- (C) -4
- (D) -1
- (E) 2
- (F) 7

13. Food spends about 4 hours in the stomach and 16 hours in the intestines, making a 20 hour journey. What fraction of the whole journey is spent in the stomach? [\[Nutrition in Animals\]](#)

- (A) $\frac{1}{20}$
- (B) $\frac{1}{2}$
- (C) $\frac{1}{5}$
- (D) $\frac{1}{16}$
- (E) $\frac{1}{6}$
- (F) $\frac{4}{5}$
- (G) $\frac{1}{4}$

14. An equilateral triangle has a certain number of lines of symmetry and a certain order of rotational symmetry. The sum of these two numbers is? [\[Symmetry\]](#)

- (A) 6
- (B) 3
- (C) 4
- (D) 9
- (E) 2
- (F) 12
- (G) 5

15. A forest canopy intercepts 25 percent of the rain that falls on it, and the rest reaches the ground. Of 80 centimetres of rainfall, how much reaches the forest floor? [\[Forests\]](#)

- (A) 55 cm
- (B) 75 cm
- (C) 20 cm
- (D) 105 cm
- (E) 100 cm
- (F) 40 cm
- (G) 60 cm

16. Add the expressions $2x$ plus 3 and $5x$ minus 7, then find the value of the sum when x equals 2. [\[Algebraic Expressions\]](#)

- (A) 7
- (B) 24
- (C) 12
- (D) -4
- (E) 18
- (F) 10
- (G) 14

17. An adult human has 32 teeth in all, of which 8 are sharp cutting incisors. What percentage of the teeth are incisors? [\[Nutrition in Animals\]](#)

- (A) 16%
- (B) 32%
- (C) 12.5%
- (D) 8%
- (E) 75%
- (F) 40%
- (G) 25%

18. A design returns to its starting look after every rotation of 72 degrees. Its order of rotational symmetry is? [\[Symmetry\]](#)

- (A) 10
- (B) 4
- (C) 3
- (D) 6
- (E) 5
- (F) 72
- (G) 8

19. In a patch of forest the number of trees to shrubs is in the ratio 5 : 3. If there are 320 plants in all, how many are trees? [\[Forests\]](#)

- (A) 180
- (B) 100
- (C) 120
- (D) 320
- (E) 200
- (F) 240
- (G) 160

20. Find the value of the expression $2x^2 + 3xy$ when x equals 2 and y equals 3. [\[Algebraic Expressions\]](#)

- (A) 22
- (B) 50
- (C) 18
- (D) 26
- (E) 34
- (F) 30

21. A meal supplies 2000 kilojoules of energy, but 15 percent of it leaves the body undigested as waste. The energy actually absorbed is? [\[Nutrition in Animals\]](#)

- (A) 1850 kJ
- (B) 1700 kJ
- (C) 300 kJ
- (D) 1300 kJ
- (E) 1985 kJ
- (F) 2300 kJ
- (G) 85 kJ

22. A regular polygon has exactly 5 lines of symmetry. Its smallest angle of rotational symmetry is?

[\[Symmetry\]](#)

- (A) 5 deg
- (B) 144 deg
- (C) 108 deg
- (D) 72 deg
- (E) 90 deg
- (F) 60 deg
- (G) 36 deg

23. A forest's area shrinks from 1200 hectares to 900 hectares after clearing. The percentage decrease in the forest area is? [\[Forests\]](#)

- (A) 33.3%
- (B) 30%
- (C) 25%
- (D) 75%
- (E) 15%
- (F) 300%
- (G) 20%

24. A rectangle has length $2x + 1$ and width x . Its perimeter simplifies to $6x + 2$. What is the perimeter when x equals 5? [\[Algebraic Expressions\]](#)

- (A) 36
- (B) 26
- (C) 32
- (D) 22
- (E) 60
- (F) 16

- 25.** The liver makes about 800 millilitres of bile each day to help digest fat. Over 5 days, how many litres of bile is produced? [\[Nutrition in Animals\]](#)
- (A) 40 L
 - (B) 8 L
 - (C) 160 L
 - (D) 4 L
 - (E) 0.4 L
 - (F) 2 L
 - (G) 4000 L
- 26.** A rectangle that is not a square has a certain number of lines of symmetry and a certain order of rotational symmetry. The product of these two numbers is? [\[Symmetry\]](#)
- (A) 8
 - (B) 6
 - (C) 0
 - (D) 4
 - (E) 16
 - (F) 2
- 27.** Decomposers break down 60 kilograms of dead plant matter and turn 45 kilograms of it into humus. What percentage of the dead matter became humus? [\[Forests\]](#)
- (A) 25%
 - (B) 60%
 - (C) 45%
 - (D) 15%
 - (E) 75%
 - (F) 133%
- 28.** Subtract the expression $3a$ minus 2 from the expression $5a$ plus 4, then find the value of the result when a equals 4. [\[Algebraic Expressions\]](#)
- (A) 14
 - (B) 10
 - (C) 2
 - (D) 8
 - (E) 6
 - (F) 18
 - (G) 20
- 29.** A person eats food worth 2500 kilocalories in a day, and breakfast supplies 20 percent of this. How many kilocalories does breakfast provide? [\[Nutrition in Animals\]](#)
- (A) 750 kcal
 - (B) 500 kcal
 - (C) 2520 kcal
 - (D) 125 kcal
 - (E) 250 kcal
 - (F) 100 kcal
 - (G) 2000 kcal

30. The capital letter H, drawn in its usual upright form, has how many lines of symmetry? [\[Symmetry\]](#)

- (A) 4
- (B) 2
- (C) 0
- (D) 3
- (E) 6
- (F) 8
- (G) 1

31. A single tree absorbs about 22 kilograms of carbon dioxide in a year. A row of 50 such trees absorbs how much carbon dioxide in one year? [\[Forests\]](#)

- (A) 72 kg
- (B) 550 kg
- (C) 110 kg
- (D) 2200 kg
- (E) 440 kg
- (F) 11000 kg
- (G) 1100 kg

32. How many terms are there in the algebraic expression $3x^2y - 2xy + 7$? [\[Algebraic Expressions\]](#)

- (A) 7
- (B) 3
- (C) 1
- (D) 4
- (E) 5
- (F) 2
- (G) 6

33. The complete human food canal, from mouth to the far end, measures about 9 metres. Written in centimetres, that length is? [\[Nutrition in Animals\]](#)

- (A) 900 cm
- (B) 990 cm
- (C) 9000 cm
- (D) 9 cm
- (E) 0.09 cm
- (F) 450 cm
- (G) 90 cm

34. A square has a number of lines of symmetry and an order of rotational symmetry. The product of these two numbers is? [\[Symmetry\]](#)

- (A) 20
- (B) 16
- (C) 4
- (D) 2
- (E) 8
- (F) 64
- (G) 12

- 35.** A forest is home to 800 deer, and their number rises by 15 percent in a good year. The deer population after the rise is? [\[Forests\]](#)
- (A) 9200
 - (B) 815
 - (C) 920
 - (D) 680
 - (E) 880
 - (F) 120
- 36.** Find the value of the expression $a^2 + 2ab + b^2$ when a equals 3 and b equals 2. [\[Algebraic Expressions\]](#)
- (A) 36
 - (B) 21
 - (C) 25
 - (D) 11
 - (E) 29
 - (F) 19
 - (G) 13
- 37.** A cow eats 40 kilograms of grass in a day, but only 65 percent of it is actually digested. The mass of grass digested is? [\[Nutrition in Animals\]](#)
- (A) 260 kg
 - (B) 65 kg
 - (C) 2.6 kg
 - (D) 14 kg
 - (E) 26 kg
 - (F) 40 kg
 - (G) 35 kg
- 38.** A pattern maps onto itself after a rotation of 40 degrees. Its order of rotational symmetry is? [\[Symmetry\]](#)
- (A) 320
 - (B) 12
 - (C) 9
 - (D) 6
 - (E) 8
 - (F) 40
- 39.** A forest's crown cover improves from 40 percent to 50 percent over ten years. By what percentage of its original value has the crown cover increased? [\[Forests\]](#)
- (A) 40%
 - (B) 125%
 - (C) 20%
 - (D) 10%
 - (E) 50%
 - (F) 12.5%
 - (G) 25%

40. Find the value of the expression $5x$ cubed when x equals 2. [\[Algebraic Expressions\]](#)

- (A) 10
- (B) 125
- (C) 15
- (D) 40
- (E) 80
- (F) 1000

41. In one day a cow's stomach makes about 1500 millilitres of digestive juice while its pancreas makes 1000 millilitres. The ratio of stomach juice to pancreas juice, simplified, is? [\[Nutrition in Animals\]](#)

- (A) 3 : 1
- (B) 3 : 2
- (C) 15 : 10
- (D) 1500 : 1000
- (E) 2 : 1
- (F) 2 : 3
- (G) 5 : 2

42. A regular polygon turns onto itself for the first time after a rotation of 30 degrees. How many sides does the polygon have? [\[Symmetry\]](#)

- (A) 24
- (B) 15
- (C) 12
- (D) 30
- (E) 6
- (F) 10
- (G) 8

43. In a simple food chain, 100 grass plants feed 10 deer, and 10 deer feed 1 tiger. A forest with 5 tigers needs how many deer to support them? [\[Forests\]](#)

- (A) 10
- (B) 5
- (C) 25
- (D) 50
- (E) 55
- (F) 500

44. Simplify the expression 4 times (x plus 3) minus 2 times (x minus 1), then find its value when x equals 3. [\[Algebraic Expressions\]](#)

- (A) 20
- (B) 18
- (C) 22
- (D) 8
- (E) 24
- (F) 14
- (G) 26

45. A 600 gram meal is 25 percent carbohydrate and 10 percent protein by mass. By how many grams does the carbohydrate exceed the protein? [\[Nutrition in Animals\]](#)
- (A) 15 grams
 - (B) 35 grams
 - (C) 150 grams
 - (D) 135 grams
 - (E) 90 grams
 - (F) 60 grams
 - (G) 210 grams
46. A semicircle (a half-disc with its straight edge) has how many lines of symmetry? [\[Symmetry\]](#)
- (A) 3
 - (B) 6
 - (C) 8
 - (D) 2
 - (E) 4
 - (F) 1
 - (G) 0
47. Only about 2 percent of the sunlight hitting the top of a thick forest reaches the floor below. If the canopy receives 5000 units of light, the floor receives about? [\[Forests\]](#)
- (A) 98 units
 - (B) 250 units
 - (C) 100 units
 - (D) 1000 units
 - (E) 2 units
 - (F) 50 units
 - (G) 200 units
48. In the expression $3y^2 - 5y + 8$, what is the coefficient of y^2 ? [\[Algebraic Expressions\]](#)
- (A) 3
 - (B) 8
 - (C) 1
 - (D) -5
 - (E) -3
 - (F) 5
 - (G) 2
49. One gram of fat releases 9 kilocalories of energy and one gram of carbohydrate releases 4 kilocalories. The energy from 10 grams of fat plus 20 grams of carbohydrate is? [\[Nutrition in Animals\]](#)
- (A) 80 kcal
 - (B) 170 kcal
 - (C) 100 kcal
 - (D) 13 kcal
 - (E) 260 kcal
 - (F) 90 kcal

50. A scalene triangle, whose three sides are all different lengths, has how many lines of symmetry?

[Symmetry]

- (A) 0
- (B) 1
- (C) 4
- (D) 3
- (E) 2
- (F) 5

51. A 600 hectare forest has one quarter of its area as open grassland and the rest covered by trees. The area covered by trees is? [Forests]

- (A) 550 ha
- (B) 300 ha
- (C) 240 ha
- (D) 500 ha
- (E) 150 ha
- (F) 450 ha
- (G) 400 ha

52. Find the value of the expression negative 2 times x squared when x equals negative 3. [Algebraic Expressions]

- (A) -36
- (B) -12
- (C) 12
- (D) 36
- (E) -18
- (F) 18
- (G) -6

53. Swallowed food stays about 3 hours in the stomach and then 6 hours in the small intestine. How many minutes does the food spend in these two parts together? [Nutrition in Animals]

- (A) 540 minutes
- (B) 360 minutes
- (C) 180 minutes
- (D) 54 minutes
- (E) 600 minutes
- (F) 90 minutes

54. A regular pentagon has a smallest angle of rotation that maps it onto itself. After three such smallest rotations in a row, the total angle turned is? [Symmetry]

- (A) 216 deg
- (B) 144 deg
- (C) 72 deg
- (D) 360 deg
- (E) 180 deg
- (F) 108 deg
- (G) 240 deg

- 55.** One hectare of a certain forest holds about 400 trees. A protected patch measuring 6.5 hectares holds about how many trees? [\[Forests\]](#)
- (A) 65
 - (B) 2400
 - (C) 260
 - (D) 1300
 - (E) 2600
 - (F) 406
 - (G) 26000
- 56.** The sum of three consecutive whole numbers can be written as $3n$ plus 3, where n is the smallest. What is this sum when n equals 10? [\[Algebraic Expressions\]](#)
- (A) 30
 - (B) 33
 - (C) 13
 - (D) 31
 - (E) 39
 - (F) 36
 - (G) 303
- 57.** A cow chews each mouthful of cud about 40 times before swallowing it again. For 30 such mouthfuls, the total number of chews is? [\[Nutrition in Animals\]](#)
- (A) 400 chews
 - (B) 70 chews
 - (C) 120 chews
 - (D) 12000 chews
 - (E) 1200 chews
 - (F) 800 chews
- 58.** If a certain regular polygon first looks unchanged only after being rotated by 24 degrees, the number of sides it has is? [\[Symmetry\]](#)
- (A) 24
 - (B) 18
 - (C) 10
 - (D) 15
 - (E) 20
 - (F) 8
- 59.** Each year about 8 tonnes of leaves fall on every hectare of a forest. Over 3 years, a 4 hectare grove receives how many tonnes of fallen leaves? [\[Forests\]](#)
- (A) 12 tonnes
 - (B) 24 tonnes
 - (C) 96 tonnes
 - (D) 15 tonnes
 - (E) 48 tonnes
 - (F) 960 tonnes
 - (G) 32 tonnes

60. The cost of buying x pens at 5 rupees each and y pencils at 2 rupees each is $5x$ plus $2y$ rupees. Find the cost when x equals 4 and y equals 6. [\[Algebraic Expressions\]](#)

- (A) 28 rupees
- (B) 32 rupees
- (C) 12 rupees
- (D) 42 rupees
- (E) 56 rupees
- (F) 20 rupees
- (G) 26 rupees

61. An amoeba traps food by surrounding it with finger-like extensions. If it captures 5 food particles every minute, how many will it have captured in 12 minutes? [\[Nutrition in Animals\]](#)

- (A) 600 particles
- (B) 72 particles
- (C) 50 particles
- (D) 17 particles
- (E) 60 particles
- (F) 120 particles

62. A regular polygon with twelve sides is turned about its centre. The smallest angle of rotation that maps it onto itself is? [\[Symmetry\]](#)

- (A) 30 deg
- (B) 60 deg
- (C) 24 deg
- (D) 45 deg
- (E) 36 deg
- (F) 12 deg
- (G) 15 deg

63. Of the 120 centimetres of rain that falls on a forest in a year, 35 percent soaks into the soil. The depth of rain that soaks in is? [\[Forests\]](#)

- (A) 78 cm
- (B) 420 cm
- (C) 35 cm
- (D) 60 cm
- (E) 30 cm
- (F) 85 cm
- (G) 42 cm

64. Find the value of the expression x squared minus y squared when x equals 7 and y equals 3. [\[Algebraic Expressions\]](#)

- (A) 40
- (B) 10
- (C) 52
- (D) 4
- (E) 20
- (F) 16
- (G) 58

65. Of the 2500 millilitres of water a person loses in a day, 60 percent leaves the body as urine. The volume lost as urine is? [\[Nutrition in Animals\]](#)

- (A) 2440 mL
- (B) 1500 mL
- (C) 150 mL
- (D) 600 mL
- (E) 1000 mL
- (F) 250 mL

66. The capital letter S, written in its usual form, has order of rotational symmetry equal to? [\[Symmetry\]](#)

- (A) 1
- (B) 2
- (C) 6
- (D) 4
- (E) 3
- (F) 0
- (G) 8

67. In a forest the animals are herbivores and carnivores in the ratio 9 : 1. Out of 500 animals, how many are carnivores? [\[Forests\]](#)

- (A) 90
- (B) 50
- (C) 100
- (D) 5
- (E) 450
- (F) 55
- (G) 250

68. Simplify the expression $6m$ minus the quantity $2m$ minus the quantity m minus 3, removing the brackets carefully, then find its value when m equals 4. [\[Algebraic Expressions\]](#)

- (A) 29
- (B) 23
- (C) 13
- (D) 21
- (E) 17
- (F) 11
- (G) 20

69. Among an adult's 32 teeth there are 8 incisors, 4 canines, 8 premolars and 12 molars. The molars make up what fraction of all the teeth, in simplest form? [\[Nutrition in Animals\]](#)

- (A) $\frac{1}{3}$
- (B) $\frac{3}{4}$
- (C) $\frac{5}{8}$
- (D) $\frac{1}{4}$
- (E) $\frac{3}{8}$
- (F) $\frac{2}{3}$
- (G) $\frac{12}{20}$

70. A figure has rotational symmetry of order 8. If it is rotated three times by its smallest such angle, the total angle through which it has turned is? [\[Symmetry\]](#)

- (A) 135 deg
- (B) 360 deg
- (C) 90 deg
- (D) 45 deg
- (E) 120 deg
- (F) 180 deg

71. A spreading banyan tree's circular canopy has a diameter of 20 metres. Taking the value of pi as 3.14, the ground area shaded by the canopy is about? [\[Forests\]](#)

- (A) 314 m²
- (B) 100 m²
- (C) 1256 m²
- (D) 62.8 m²
- (E) 628 m²
- (F) 31.4 m²

72. Which of these is a like term that can be combined with $7xy$? [\[Algebraic Expressions\]](#)

- (A) $2xy$ squared
- (B) $7y$
- (C) $5x$
- (D) $3yx$
- (E) $7x$ squared y
- (F) $7x$

73. A growing child needs about 1800 kilocalories a day while an adult needs 2400 kilocalories. The adult needs what percentage more energy than the child? [\[Nutrition in Animals\]](#)

- (A) 33.3%
- (B) 75%
- (C) 600%
- (D) 30%
- (E) 133%
- (F) 20%
- (G) 25%

74. An isosceles triangle, with exactly two equal sides, has how many lines of symmetry? [\[Symmetry\]](#)

- (A) 5
- (B) 1
- (C) 6
- (D) 3
- (E) 2
- (F) 4
- (G) 0

- 75.** A young forest tree grows taller by about 0.5 metres each year. Growing steadily, how tall will it be after 30 years? [\[Forests\]](#)
- (A) 20 m
 - (B) 6 m
 - (C) 1.5 m
 - (D) 30.5 m
 - (E) 150 m
 - (F) 15 m
 - (G) 60 m
- 76.** Find the value of the expression $2a^3 - a$ when a equals 3. [\[Algebraic Expressions\]](#)
- (A) 15
 - (B) 45
 - (C) 54
 - (D) 51
 - (E) 23
 - (F) 57
 - (G) 48
- 77.** A human tongue carries about 10000 taste buds, and roughly 25 percent of them respond to bitter tastes. The number of bitter-sensing taste buds is about? [\[Nutrition in Animals\]](#)
- (A) 5000
 - (B) 7500
 - (C) 40
 - (D) 2500
 - (E) 1000
 - (F) 400
 - (G) 250
- 78.** A rhombus that is not a square has its lines of symmetry along its diagonals. How many lines of symmetry does it have? [\[Symmetry\]](#)
- (A) 1
 - (B) 2
 - (C) 4
 - (D) 3
 - (E) 8
 - (F) 0
- 79.** A district covers 8000 square kilometres, of which 24 percent is under forest. The forest area of the district is? [\[Forests\]](#)
- (A) 2400 km²
 - (B) 192 km²
 - (C) 480 km²
 - (D) 6080 km²
 - (E) 1600 km²
 - (F) 19200 km²
 - (G) 1920 km²

- 80.** A rectangle has length x plus 2 and width x , so its area is x squared plus $2x$. What is the area when x equals 4? [\[Algebraic Expressions\]](#)
- (A) 16
 - (B) 48
 - (C) 20
 - (D) 24
 - (E) 32
 - (F) 28
- 81.** A dairy cow gives 12 litres of milk each day. Across the 30 days of June, the total milk it gives is? [\[Nutrition in Animals\]](#)
- (A) 300 L
 - (B) 360 L
 - (C) 3600 L
 - (D) 42 L
 - (E) 36 L
 - (F) 120 L
 - (G) 180 L
- 82.** A regular polygon has 9 lines of symmetry. Its smallest angle of rotational symmetry is? [\[Symmetry\]](#)
- (A) 36 deg
 - (B) 72 deg
 - (C) 90 deg
 - (D) 40 deg
 - (E) 20 deg
 - (F) 45 deg
 - (G) 9 deg
- 83.** Decomposers clear about 2 kilograms of leaf litter from a plot every day. At this steady rate, a 50 kilogram pile of litter is cleared in how many days? [\[Forests\]](#)
- (A) 12.5 days
 - (B) 52 days
 - (C) 10 days
 - (D) 100 days
 - (E) 25 days
 - (F) 30 days
 - (G) 48 days
- 84.** Find the value of the expression $(3x)$ squared when x equals 2. [\[Algebraic Expressions\]](#)
- (A) 6
 - (B) 12
 - (C) 24
 - (D) 36
 - (E) 9
 - (F) 18
 - (G) 72

85. The food pipe (oesophagus) is about 25 centimetres long while the small intestine is about 750 centimetres long. The small intestine is how many times the length of the food pipe? [\[Nutrition in Animals\]](#)

- (A) 15 times
- (B) 20 times
- (C) 300 times
- (D) 25 times
- (E) 3 times
- (F) 30 times
- (G) 725 times

86. A regular polygon with ten equal sides is turned about its centre. Its smallest angle of rotational symmetry is? [\[Symmetry\]](#)

- (A) 30 deg
- (B) 10 deg
- (C) 60 deg
- (D) 18 deg
- (E) 45 deg
- (F) 72 deg
- (G) 36 deg

87. A healthy forest stores about 150 tonnes of carbon in every hectare. A protected reserve of 20 hectares stores how much carbon? [\[Forests\]](#)

- (A) 170 tonnes
- (B) 7.5 tonnes
- (C) 1500 tonnes
- (D) 30000 tonnes
- (E) 300 tonnes
- (F) 750 tonnes
- (G) 3000 tonnes

88. Simplify the expression $2x^2 + 3x^2 - x$ by combining like terms, then find its value when x equals 2. [\[Algebraic Expressions\]](#)

- (A) 8
- (B) 10
- (C) 18
- (D) 16
- (E) 14
- (F) 38
- (G) 22

89. A balanced plate is planned as 50 percent grains, 30 percent vegetables and 20 percent protein by mass. On a 400 gram plate, the mass of vegetables is? [\[Nutrition in Animals\]](#)

- (A) 120 g
- (B) 200 g
- (C) 80 g
- (D) 60 g
- (E) 240 g
- (F) 30 g

- 90.** Consider a slanted four-sided shape whose opposite sides are parallel but which is neither a rectangle nor a rhombus. The number of lines of symmetry it has is? [\[Symmetry\]](#)
- (A) 1
 - (B) 0
 - (C) 2
 - (D) 3
 - (E) 8
 - (F) 4
- 91.** After protection, the number of bird species in a forest rises from 80 to 100. The percentage increase in the number of species is? [\[Forests\]](#)
- (A) 18%
 - (B) 125%
 - (C) 80%
 - (D) 25%
 - (E) 40%
 - (F) 20%
 - (G) 30%
- 92.** A rectangle has length $3a$ and width a , so its perimeter is $8a$. What is the perimeter when a equals 5? [\[Algebraic Expressions\]](#)
- (A) 20
 - (B) 45
 - (C) 16
 - (D) 30
 - (E) 35
 - (F) 80
 - (G) 40
- 93.** After digestion of a 250 gram meal, 40 grams leaves the body as undigested waste. What percentage of the meal was absorbed? [\[Nutrition in Animals\]](#)
- (A) 21%
 - (B) 84%
 - (C) 60%
 - (D) 16%
 - (E) 210%
 - (F) 75%
 - (G) 40%
- 94.** A star design has rotational symmetry of order 5. If it is rotated twice by its smallest such angle, the total angle turned is? [\[Symmetry\]](#)
- (A) 108 deg
 - (B) 360 deg
 - (C) 144 deg
 - (D) 72 deg
 - (E) 90 deg
 - (F) 288 deg
 - (G) 216 deg

95. A forest fire sweeps through $\frac{3}{5}$ of a 250 hectare woodland. The area of woodland left untouched is?

[Forests]

- (A) 50 ha
- (B) 100 ha
- (C) 150 ha
- (D) 75 ha
- (E) 125 ha
- (F) 200 ha
- (G) 60 ha

96. Find the value of the expression x divided by 2 plus x divided by 3 when x equals 12. [Algebraic Expressions]

- (A) 5
- (B) 2.4
- (C) 6
- (D) 8
- (E) 12
- (F) 10
- (G) 24

97. A python swallows prey that weighs 8 kilograms, which is 40 percent of the python's own body mass. The python's body mass is? [Nutrition in Animals]

- (A) 12 kg
- (B) 3.2 kg
- (C) 20 kg
- (D) 48 kg
- (E) 5 kg
- (F) 16 kg

98. The smallest turn that brings a certain regular polygon back to its original look is 18 degrees. How many sides does the polygon have? [Symmetry]

- (A) 40
- (B) 20
- (C) 36
- (D) 18
- (E) 24
- (F) 15
- (G) 10

99. A single wild tiger needs a hunting territory of about 60 square kilometres. A forest reserve of 480 square kilometres can support how many tigers without their ranges overlapping? [Forests]

- (A) 420
- (B) 540
- (C) 28800
- (D) 10
- (E) 6
- (F) 12
- (G) 8

100. Simplify the expression 2 times (x plus 3) plus 3 times (x minus 1) by expanding and collecting like terms, then find its value when x equals 4. [\[Algebraic Expressions\]](#)

- (A) 11
- (B) 35
- (C) 25
- (D) 23
- (E) 29
- (F) 17